

Review

We thank the editor and reviewers for the positive feedback and thorough comments. Changes are in **bold** within this document and the revised manuscript.

Editor

All three reviewers noted that the submission would benefit from additional justification of the proposed methods, as well as a more accessible description of various analyses and corresponding benchmarks for interpretation. Examples are provided by reviewers and are evident throughout the paper. As noted in the text, there are benefits to alternative approaches to measure development and validity testing; the challenge is making sure that readers are clear on the relevant procedures, what advantages – specifically – the proposed methods hold over conventional approaches, and how corresponding results (models, estimates, tables, graphs) are to be interpreted.

We sincerely thank the Editor for the positive feedback, the thorough evaluation of our manuscript, and the opportunity to submit a revision. Your guidance regarding the justification of our methods and the description of our samples has been invaluable in improving the accessibility and overall quality of the paper.

In their comments, Reviewer 2 provides some suggestions for testing across different demographic characteristics. I would also note that samples are not well described in the current submission, making it difficult to gauge the potential generalizability of results. Existing research suggests that cultural factors may influence how respondents experience and report somatic phenomena, making this an important consideration for the project. A more comprehensive description of participants is needed as well as a meaningful discussion of any relevant limitations/implications for interpretation (the discussion of the proposed Olfa and Dem clusters on pg. 41 is a thoughtful example).

We have now added more information on the participants. Check lines 191-195 for study 1 participants and lines 393-395 for study 2 participants.

Study 1:

The final sample includes 559 participants [...] 50.8% women, **1.07% non-binary**; country of residence: 63.86% UK, 26.65% USA, 9.48% other). **Within this sample, 68.3% identified as White, 11.3% as Black, 5.7% as Mixed ethnicity, and 4.1% as South Asian, 9.3% as others and 0.7% preferred to not disclose.**

Study 2:

The final sample includes 737 participants [...] 57.3% women, **1.63% non-binary**; Country of residence: 75.17% United Kingdom, 24.83% other).

Within this sample, 80.3% identified as White, 6.6% as Black, 4.9% as Mixed ethnicity, and 3.0% as South Asian and 4.3% as others.

In terms of discussing relevant limitations/implications of sample characteristics on the Mint structure, we have added the following to the discussion section (lines 764-779):

“A further limitation concerns the predominantly Western, UK-based composition of the samples, which restricts the extent to which the Mint’s structure can be assumed to generalise cross-culturally, particularly given that interoceptive processes, and their cognitive elaborations, vary according to demographic factors such as gender/sex and culture (e.g., Ma-Kellams et al., 2024). For example, the Japanese validation of the MAIA-2 (Shoji et al., 2018) identified a six-factor structure rather than the original eight-factor model, partly due to the removal of items from the Self-Regulation and Not-Worrying dimensions, reflecting conceptual and translation-related differences. Similarly, (Freedman et al., 2021) reported that Japanese participants required greater contextual specification of items, highlighting cultural variation in how bodily experiences are interpreted. Language-related constraints have also been observed in cross-cultural scale adaptation, such as item-level ambiguities in (Lin et al., 2023) IAS validation. These findings suggest that the MINT’s metacluster organisation may partly reflect Western norms of bodily reporting. Future research should therefore formally test measurement invariance across culturally diverse samples to determine whether the identified clusters are structurally stable or culturally contingent. While the overall [...].”

Reviewer 1 raises important questions about the interpretation and labeling of multiple response clusters. This is a critical issue that extends to the measure as a whole. While there may be no unified model of interoception, there is a robust literature in this area that should serve as a foundation for the interpretation of results (again the discussion of proposed Olfa and Dem clusters on pg. 41 provides a good example). A more comprehensive description of the identified clusters, the proposed hierarchical structure, and connections of the preferred model to current theory and existing measures is needed. A structured set of predictions/hypotheses

specifying how external scales and participant characteristics should relate to proposed cluster scores will also be important for guiding interpretations.

We agree that clearer theoretical interpretation of the identified clusters is important. We have therefore expanded the manuscript to provide a more detailed description of the higher-order and lower-order clusters.

Lines 573-592:

“The final structure included 11 lower-level [...]. The “Interoceptive Awareness” cluster refers to self-perceived awareness of specific bodily states and sensations, particularly those associated with distinct physiological conditions. In the present paper, we do not conceptualise awareness as the correspondence between behavioural measures of interoceptive accuracy and metacognitive insight, as proposed in the model of Garfinkel et al. (2015); instead, we use the term more broadly to refer to general bodily states representations accessible to self-report. The items in this cluster therefore capture perceived awareness of bodily states such as relaxation (RelA), sensations associated with sexual arousal (SexS), and impending expulsive events (ExAc; e.g., burping, sneezing, or flatulence), reflecting an individual’s perceived ability to recognise these bodily states. The “Interoceptive Deficit” cluster reflects difficulties, uncertainty, or inaccuracies in recognising internal bodily signals. Items in this cluster capture experiences such as confusion about cardiorespiratory sensations (CaCo), uncertainty about urinary or gastrointestinal needs (UrIn), delayed recognition of hunger or thirst (Sati), contribution of external cues such as checking body odours (Olfa), and heightened sensitivity to dermal stimulation (Derm). Finally, “Visceroception” corresponds broadly to the concept of interoceptive sensibility described in Khalsa et al. (2018) model as the self-reported tendency to notice internal bodily sensations. In the present scale, this construct is specifically focused on sensations arising from three major visceral systems (i.e., cardiac, respiratory, and gastric). Accordingly, items assess perceived sensitivity to changes in heart activity (Card), breathing (Resp), and stomach sensations (Gast).”

We agree that specifying theoretically grounded predictions regarding how the identified clusters should relate to external constructs is important for guiding interpretation and future research. However, given the exploratory and data-driven nature of the present work, we believe that a more comprehensive set of theoretically derived predictions will be better developed and tested in subsequent studies.

In particular, an important next step will be to investigate potential clusters of participants within the multidimensional space defined by the Mint dimensions. While the present study focused on identifying the structure among variables (i.e.,

clusters of items and dimensions), it remains an open question whether individuals themselves form distinct interoceptive profiles or whether variation across these dimensions is more continuous. To highlight this direction, we have added the following statement to the future directions section of the manuscript (lines 744-747):

“Overall, the Mint provides a more specific measure of self-reported interoceptive processing across contexts and modalities, offering researchers a granular and comprehensive tool with which to further understand the contribution of bodily awareness to human health and disease. **In particular, the multidimensional structure of the Mint may allow future work to identify distinct interoceptive profiles, rather than assuming that individuals vary along a single continuum of interoceptive ability.**”

At the same time, we considered that the reviewer might be referring to testable hypotheses regarding the relationships between the identified clusters and external measures included in Study 2. To clarify this point, we have therefore added explicit predictions regarding the expected associations between the Mint metaclusters and theoretically related constructs (lines 377-386):

“Finally, we assessed the Mint’s criterion validity by directly comparing the predictive power of all interoception questionnaires across a broad range of outcomes, including emotional, cognitive, and clinical variables. **We hypothesised that if the *Interoceptive Deficit* metacluster reflects difficulties recognising bodily signals, it should correlate positively with constructs reflecting impaired bodily and emotional awareness, such as alexithymia and dissociative experiences, and be elevated in conditions characterised by altered interoceptive processing (e.g., autism or ADHD). In contrast, *Visceroception*, capturing perceived sensitivity to cardiac, respiratory, and gastric signals, should relate more strongly to measures of bodily noticing and physiological monitoring behaviours, including the Noticing dimension of the MAIA-2. Finally, *Interoceptive Awareness*, which captures perceived access to specific bodily states such as relaxation or sexual arousal, should relate more strongly to measures assessing conscious awareness.**”

We hope that these additions provide a clearer interpretative framework for the Mint dimensions and offer a basis for future research investigating both theoretical predictions about interoceptive constructs and potential clusters of individuals within the multidimensional space defined by the scale.

Reviewer 2 expressed a number of valid concerns regarding item coverage in the initial pool and the decision to alter the set of indicators submitted for analysis in Study 2. Continued refinement across samples is expected, but Study 2 essentially

tests a new scale which, in the absence of a subsequent replication sample, undermines conclusions about the stability and generalizability of responses. It is difficult to know based on the information provided whether the 33-item scale is a replicable solution or better viewed as a sample-specific model. This is a serious question that may be difficult to address with the data available. Separating samples into test and replication sets could provide a solution.

We agree with the reviewer and editor that not having a replication sample for the final 33 items scale undermines overall conclusions. To address this, we conducted EGA in an independent third sample (N = 359, Mean age = 41.9, SD =12.6, range [20,82], 51% women and 0.56% non-binary; 73.26% United Kingdom, 22.01% United States, 4.74% other) collected as part of an unrelated unpublished study (data available at <https://github.com/RealityBending/FakeArt/tree/main/data>), using only the final 33 items.

In this new sample, the lower-order clusters organised into three-item clusters, consistent with the EGA solution reported in the manuscript. Item stability was generally acceptable (often exceeding 0.70), although three of the 33 items showed notably lower stability (0.25–0.30).

At the higher-order level, and consistent with the manuscript EGA, items are organised into three higher-order clusters. Replicating the manuscript findings, the new sample also produced the same Visceroception cluster (Respiration, Gastroception, Cardioception).

In the manuscript EGA, three lower-order clusters comprised the Interoceptive Awareness metacluster. In the new sample, the same metacluster emerged, with the addition of Dermal Hypersensitivity.

In sum, the higher-order structure was identical across samples, with the only difference being that the Dermal cluster shifted from Interoceptive Deficit to Interoceptive Awareness. We believe this additional analysis strengthens confidence in the robustness of the proposed structure, while also reinforcing our interpretation that the dermal and olfactory lower-level clusters may be comparatively less stable and warrant further investigation.

While we believe that this is useful information for the reviewers, we would prefer not include this preliminary extra data in the manuscript, given that the data collection for this project is still on-going, and that integrating it would require inserting a substantial amount of information to describe the new sample and context.

As noted by Reviewer 2, the number and complexity of analyses included in the submission are sometimes difficult to follow and may push the limits of what can be expected from a single paper and/or a given sample (even those with a large number of observations). There are several aspects of the approach to development and validity testing that could be seen as more periphery to the primary aims of the paper (the simultaneous effort to test short-form versions of the scale is one example). Streamlining the submission would help to focus the project while creating room to address conceptual, methodological, and interpretive issues raised by reviewers

We have carefully evaluated the opportunities for streamlining, and tried our best to efficiently present results that we believe are worthy of interest. While we agree that some aspects might appear somewhat tangential to the core purpose of the study, such as the exploratory testing of shorter versions, we tried to best take into consideration the “usefulness vs bloat” tradeoff, and in this particular case, would like to suggest to keep it, as we believe these results be warranted given the general trend - for better or worse - towards brief versions in the context of online studies.

.Minor Points:

- It is difficult, at present, to gauge the adherence of the paper to parameters set by the journal. Compiling tables, figures, and captions at the end of the submission would be helpful.

We have now added the tables, figures and their captions at the end of the submission (i.e., after references).

- Please remember that reliability and validity are best conceptualized as properties of interpretation as opposed to characteristics of tests themselves. Reference to “validated” scales should be avoided.

We have now changed any instance in which we describe the Mint as “validated” or being validated. We have instead referenced its convergent and criterion validity where appropriate.

Reviewer 1

Comments to the Author

The manuscript presents the development and evaluation of a new multimodal interoception questionnaire (Mint). The development was based on the modality-by-context idea, which provided an agnostic, data-driven approach to the construct. The paper is well-written, theoretically founded, and the empirical part is also ok. I just have a few comments and remarks. I'm not saying that I'm familiar

with all statistical methods applied here; therefore, I focus more on theoretical-conceptual issues.

We thank Reviewer 1 for their thoughtful evaluation, their positive remarks regarding the theoretical foundation of our work, and their highly constructive conceptual feedback. We have addressed your comments in detail below.

Introduction

"Among the numerous interoceptive questionnaires available, a handful dominate the literature" - please, also explicitly mention questionnaires assessing body awareness, an overlapping construct (for a review, see Mehling et al., 2009). From a broader point of view, other questionnaires (perhaps most importantly the BAQ) are also worthy of mentioning as it is psychometrically sound and widely used.

We agree that the BAQ is a psychometrically sound measure that remains widely used in interoception research. For this reason, we have now included the BAQ as an example of a scale that has been particularly influential in the literature. However, given the length concern highlighted by the reviewer, we aimed to avoid overloading the manuscript with references to additional measures, despite acknowledging their continued relevance.

Lines 56-66:

"Among the numerous interoceptive questionnaires available, a handful dominate the literature [...] **the Body Awareness Questionnaire (BAQ, Shields et al., 1989), [...]. In contrast, the BAQ was developed to address limitations of earlier measures by assessing beliefs about one's sensitivity to normal, non-emotive bodily processes, explicitly targeting the domain of reported awareness of normal bodily processes not typically associated with emotion or somatic complaint. The MAIA [...]. More recently, the IAS was recently developed to circumvent these high-level - possibly tangential - constructs, instead capturing subjective reports of concrete bodily events (e.g., coughing, sneezing) as proxies for interoceptive accuracy.**

Lines 72-77:

Other questionnaires include the Body Consciousness Questionnaire (BCS, Miller et al., 1981), the Self-Awareness Questionnaire (SAQ, Longarzo et al., 2015), the Interoceptive Sensitivity and Attention Questionnaire (ISAQ, Bogaerts et al., 2022), the Interoceptive Confusion Questionnaire (ICQ, Brewer et al., 2016) and the Interoceptive Attention Scale (IATS, Gabriele et al., 2022). For an extensive review of measures used in the field of interoception see Desmedt et al. (2022) and Mehling et al. (2009)."

*For example, interoceptive distinctions that are relevant at a neuroanatomical or cognitive level might not necessarily map onto the structure of subjective experiences." - I could not agree more. Perhaps it would be important to elaborate on this idea. What we perceive is a unitary body, the end-product of multisensory integration of bottom-up information, heavily influenced by top-down factors.

We have now expanded on this point by adding the following:

Lines 87-89:

For example, interoceptive distinctions that are relevant at a neuroanatomical or cognitive level might not necessarily map onto the structure of subjective experiences, **because the body constitutes a unitary representation produced by multisensory integration of bottom-up information, with substantial influence from top-down factors (Kirsch & Kunde, 2023).**

Study 1

"This systematic framework yielded a diverse pool of 120 initial items"; also (later) "we generated 120 initial items, striving for a balanced number of items with consistent phrasing within modalities and facets"- It is not clear how exactly those items were generated.

We appreciate the feedback and agree that more information on the generation of items is needed, therefore we have written the following to clarify that items were generated through continuous communication between the authors.

Lines 150-167:

"Item development proceeded through a structured process aimed at maximising consistency and breadth across interoceptive modalities and experiential contexts. Existing interoception questionnaires were first systematically reviewed to characterise the current measurement landscape, including content scope, item phrasing, and response formats. Insights from this stage, together with relevant theoretical literature and expert domain knowledge, informed the construction of a modality-by-context grid (see item generation section below), which was designed to capture and structure core principles of interoceptive experience while identifying gaps in existing measures. Items were then generated to populate each cell of this grid, using simple, first-person statements with a consistent grammatical structure. Where relevant, item wording drew inspiration from existing instruments; however, all items were newly formulated to explicitly specify both the bodily modality and the experiential context. Items were iteratively refined through internal discussion to ensure clarity, uniformity of phrasing within facets, and balanced representation across modalities and contexts. Throughout this process, redundant or ambiguous items were revised or removed, and additional items were

generated where coverage was deemed insufficient. Importantly, item generation was not guided by anticipated latent dimensions or clustering, but rather by the goal of comprehensive and conceptually coherent coverage of objectively defined modalities and contexts."

Study 2

- Please, report the internal consistency values for the questionnaires (MAIA-2 scales, BPQ, etc.) and also for the new scale

We have now reported the internal consistency in the study 2 for all relevant questionnaires. The following was added to the manuscript.

Lines 403-405:

The MINT [...]. **Correspondingly, the high-order meta-clusters of the mint showed good to excellent internal consistency: Interoceptive Awareness ($\omega = 0.85$), Interoceptive Deficit ($\omega = 0.74$), Visceroception ($\omega = 0.88$).**

Lines 413- 416:

The Multidimensional Assessment of Interoceptive Awareness [...]. **The subscales of the MAIA-2 showed an acceptable to good internal consistency: ω Attention regulation = 0.87, ω Body listening = 0.76, ω Emotional awareness = 0.79, ω Not-distracting = 0.80, ω Noticing = 0.68, ω Not-worrying = 0.79, ω Self regulation = 0.84, and ω Trusting = 0.84.**

Line 419:

The Body Perception Questionnaire - Very Short Form [...]. **Internal consistency for the BPQ-Awareness was good ($\omega = 0.91$).**

Line 422:

The Interoceptive Accuracy Scale [...]. **The IAS had good internal consistency ($\omega = 0.88$).**

Lines 426-428:

The Toronto Alexithymia Scale TAS [...]. **In this study, the TAS subscales showed poor to acceptable internal consistency: ω DIF = 0.77, ω DDF = 0.53 and ω EOT = 0.69.**

Lines 433-434:

The Emotion Reactivity Scale - Brief Version [...]. **The subscales of the B-ERS showed acceptable to good internal consistency: ω Arousal = 0.76, ω Sensitivity = 0.90 and ω Persistence = 0.74.**

Lines 438-441:

The Cognitive Emotion Regulation Questionnaire [...]. **The strategies include Self-blame ($\omega = 0.78$), Other-blame ($\omega = 0.8$), Rumination ($\omega = 0.61$), Catastrophizing ($\omega = 0.84$), Putting into Perspective ($\omega = 0.70$), Positive Refocusing ($\omega = 0.80$), Positive Reappraisal ($\omega = 0.75$), Acceptance ($\omega = 0.75$), and Planning ($\omega = 0.66$). The internal consistency for these dimensions ranged from acceptable to good.**

Lines 446-449:

The Černis Felt Sense of Anomaly scale - short form [...] It includes Anomalous Experience of the Self ($\omega = 0.75$), Anomalous Experience of the Body ($\omega = 0.77$), Anomalous Experiences of Emotion ($\omega = 0.81$), Altered Sense of Familiarity ($\omega = 0.72$), Altered Sense of Connection ($\omega = 0.86$), Altered Sense of Agency ($\omega = 0.78$), and Altered Sense of Reality ($\omega = 0.58$). **The internal consistency of the subscales were good with the exception of Sense of Reality.**

Lines 451-455:

The Primal World Beliefs Inventory [...]. **In this study, the internal consistency of the subscales were all good: ω Good = 0.84, ω Safe = 0.80, ω Enticing = 0.80 and ω Alive = 0.84.**

Lines 460-461:

We used the refined version of the PHQ-4 [...]. **PhQ-Anxiety showed excellent reliability ($\omega = 0.91$) and PHQ-Depression showed good reliability ($\omega = 0.84$).**

- It is not clear to me whether the authors consider the lower-level and higher-order clusters of Mint/metaMint/etc.) as subscales or only the total score of Mint matters - please, clarify. Perhaps the scales could be described in the appendix.

We agree that clarification on this matter is needed. We consider both lower-level and higher-order clusters as subscales, and we do not recommend calculating an overall average score for the entire MINT, as it includes both positive and negative dimensions.

Instead, we recommend that researchers select the scoring approach that best aligns with their research questions. For studies seeking a more nuanced assessment of interoception, we recommend calculating the average score for each lower-level cluster. If a more general overview is sufficient, researchers may calculate scores for the higher-order clusters only. When a comprehensive assessment of interoception is desired but time constraints are a concern, we recommend using the miniMint, which derives higher-order cluster scores from the lower-level clusters that best represent them. Finally, when time is highly limited, the microMint may be used;

this version includes only nine items (three per lower-level cluster) corresponding to the lower-level clusters that best represent each higher-order cluster.

To clarify this we have added the following to the manuscript:

Lines 400-403:

“The 45 items of the Multimodal Interoception Questionnaire (Mint) scale were presented in a random order, with the same 7-point Likert scale as in study 1 (0 = Disagree, 6 = Agree). **Scores should be calculated as the average of the items within each lower-order cluster. However, when a broader summary is required, average scores for the higher-order meta-clusters can be computed from the average of all their corresponding items.** Correspondingly, the subscales of the mint showed good to excellent internal consistency: Interoceptive Awareness ($\omega = 0.85$), Interoceptive Deficit ($\omega = 0.74$), Visceroception ($\omega = 0.88$).”

The following descriptions of the scales have been included in the data analysis scripts (QMD and HTML), which will be made available following manuscript acceptance.

Interoceptive Awareness - Awareness of specific bodily states and internal sensations

- Expulsion Accuracy (ExAc) - Items describing self-perceived accurate awareness of upcoming expulsion sensations (e.g., burp, fart, sneeze).
- Relaxation Awareness (RelA) – Items describing awareness of the bodily state associated with relaxation
- Sexual Arousal Sensitivity (SexS) – Items describing awareness of sensations linked to sexual arousal.

Interoceptive Deficit - Confusion, inaccuracy, or delayed awareness of internal bodily signals

- Cardiorespiratory Confusion (CaCo) - Items describing uncertainty about heart and breathing signals
- Urointestinal Inaccuracy (UrIn) - Items describing uncertainty about urination and defecation signals
- Dermal Hypersensitivity (Derm) - Items describing heightened skin sensitivity to touch or materials
- Satiety Noticing (Sati) - Items describing delayed awareness of hunger and thirst
- Olfactory Contribution (Olfa) - items describing the tendency to check body odours to monitor bodily states

Visceroception - Sensitivity to internal visceral sensations

Interoceptive sensibility

- Respiroception (Resp) - Items describing sensitivity to changes in breathing
- Cardioception (Card) – Items describing sensitivity to changes in heart activity
- Gastroception (Gast) – Items describing sensitivity to stomach sensations

We have added a more detailed description of the high-order and low-order clusters in the manuscript; see lines 573-592. For clarification of what it entails please see answer to editor above.

- I don't really agree with the interpretation of olfaction. First, olfaction is closely linked to interoception. Second, the items do not refer to any kind of compensation - olfaction is simply used to get information that could not be acquired via other channels. In sum, those items suggest that body-related olfactory information contributes to the representation of the body, i.e., it gets integrated with other modalities. Actually, the same remark applies to respiration - all three items refer to information belonging more to proprioception than interoception in the strict sense. Exactly this is why multisensory integration should be presented in detail in the introduction.

We agree that the term compensation may not accurately reflect the function captured by these items and have therefore renamed this cluster to "Olfactory Contribution". We believe this label now better reflects the idea that olfactory information can contribute to the representation of bodily states rather than compensating for a lack of other signals, through integration with other sensory channels.

With regard to the items themselves, the Olfactory Contribution (Olfa) items capture behaviours such as checking one's own body odours (e.g., armpits, breath, or flatulence). These behaviours reflect the use of olfactory information to monitor or infer bodily states, suggesting that body-related olfactory cues may contribute to how individuals represent and interpret internal bodily conditions.

Regarding the reviewer's comment about respiration, we acknowledge that respiratory signals can involve both interoceptive and proprioceptive components. However, the items in the Respiroception (Resp) cluster focus on perceived sensitivity to internal respiratory sensations that convey information about the current state of the body which have been classified as interoceptive (Nikolova et al.,

2022). For this reason, we retained these items within the interoceptive group while clarifying in the introduction that bodily awareness likely reflects the integration of multiple sensory sources (see answer above).

- Perhaps a limitation, perhaps a necessity is that - although the authors started from a context-dependent point -, generalization remains there. In more detail, (1) many items suggest generalization ('I often', 'always', etc.); (2) only certain facet-modality combinations were kept in the final Mint. This is not a problem at all but should be discussed.

We agree on the need to further explain our use of generalisation in some items. The use of "I often" and "I always" etc, was a purposeful choice as we intend to tap into trait-like, dispositional characteristics but we agree that in doing so we might miss important aspects of which the occurrence of signals is either rare or sporadic. For the purpose of transparency and acknowledgement of this choice as a limitation we will discuss this issue further in the manuscript. We have now changed lines 755-763 in the limitation section to clarify this point.

"We note several limitations that could serve as the impetus for future research efforts. [...] **Thirdly, although the final version of the MINT provides broader coverage of interoceptive domains than many existing alternatives, it remains non-exhaustive. In particular, several theoretically relevant content areas were not retained in the final item set, including sexual arousal state awareness, sexual arousal physiological changes (e.g., cardiac, respiratory, and thermoregulatory changes), interoceptive signalling items (e.g., noticing bodily changes during personally salient events), and pain- or illness-related interoceptive sensitivity. Future work may therefore benefit from reintroducing and refining these domains to further expand the scope of the instrument, while balancing content coverage against psychometric performance and scale length. "**

Reviewer 2

Comments to the Author

The present manuscript proposes and conducts a first validation of the Multimodal Interoception Questionnaire (the Mint Scale) across 2 studies. In the first study, the authors use a bottom-up approach to item generation where modalities (e.g., cardiac, respiratory) and contexts (e.g., negative, positive) were crossed to compile the initial set of 120 questionnaire items. Network psychometrics were used to evaluate the stability and dimensionality of the resulting structure leading to an item pool of 42 items. In the second study, the authors further reduced the item pool to 33 items and evaluate both concurrent and criterion validity. Overall, I found this paper to be an interesting read with extensive analysis; however, I have several comments

for the authors which I believe will strengthen the manuscript and make it more impactful for readers.

We thank Reviewer 2 for their detailed analysis of our work and their valuable suggestions, which will undoubtedly strengthen the manuscript.

Comments:

1. The authors elect to use a bottom-up item generation approach alongside network psychometrics to construct their instrument. This is different than the traditional top-down approach to instrument development. I believe some additional discussion of the tradeoffs associated with the bottom-up approach would be beneficial.

We agree with the reviewer that characterizing our method as a strict “bottom-up” versus “top-down” dichotomy is misleading and oversimplifies the item generation process.

What we initially meant by “bottom-up” is that we did not start with a *theoretical* model of interoception (e.g., measuring interoceptive “sensitivity”, “confusion” or “attention”). We instead tried to be as wide as possible in the item generation procedure, creating a grid of modalities x contextes to force us to think of items in each category, that we might otherwise overlook as some might be somewhat untypical or incongruent (e.g., negative sensations of genitalia, or gastric changes during positive arousal). But as the reviewer suggests, it would be misleading to frame this as a pure bottom-up approach.

Moreover, scale development methods are typically a mix of top-down a priori ideas and bottom-up data-driven decisions. While our approach diverges from traditional methods by relying on network psychometric indices (such as bootstrapped stability) rather than inferential models like Confirmatory Factor Analysis (CFA), it is not a radically statistically bottom-up approach either. Our approach represents a parallel path to the same goal rather than a complete departure from traditional methods. Recognizing your concern, we have revised the manuscript to soften this framing and have removed the term 'bottom-up' to better reflect this nuanced process.

2. Similarly, since hierarchical EGA is a relatively new approach to psychometric analysis, I would encourage the authors to include appropriate benchmarks for EGA fit statistics (e.g., cluster stability) where possible.

We agree with the reviewer and have added the following information to the manuscript regarding item stability (lines 270-285)

After removing redundant items using UVA, we iteratively fitted hierarchical EGA models (which additionally estimate higher-order “meta” clusters) using “glasso” (Friedman et al., 2008) and the “Leiden” algorithm (Traag et al., 2019) for community

detection, refining the item pool at each step. **We then evaluated item stability (e.g., extent to which items shift between clusters across bootstrapped samples). Stability ranges from 0 (no stability) to 1 (perfect stability), with values below 0.75 indicating insufficient stability (Christensen & Golino, 2021). Although the recommended removal threshold is ≥ 0.70 we applied a more conservative criterion and removed items with stability below 0.80 to preserve stronger indicators from a large initial item pool. We also removed items that did not belong to any cluster and item pairs, retaining only items from clusters with more than two items. Finally, within each lower-level cluster, we selected the three items with the highest node centrality (i.e., the strongest loadings within the cluster). To assess dimension stability, we evaluated the robustness of the identified clusters using bootstrapping, which provides indices of structural consistency (i.e., the proportion of bootstrap samples in which the original dimension is exactly replicated). High stability (e.g., $>70\%$; Christensen & Golino, 2021) indicates a robust dimension, whereas lower values suggest unstable, sample-dependent, or poorly defined dimensions. Lastly, we will assess model fit using the Generalized Total Entropy Fit Index (TEFI; Golino et al., 2021), where higher values (i.e., values closer to zero) indicate better fit."**

We made minor changes following the addition of this information on the analysis section (lines 304-315):

"The final hierarchical EGA model (**TEFI = -119.18**), in which all 54 items yielded a high cluster stability ($> 90\%$) suggested 3 metaclusters and 15 lower-level clusters (each containing 3 items).[...] **Of note, the dimension stability of the higher clusters showed low structure consistency for all high-order clusters (23%-47%)."**

Line 545:

"The application of EGA to the initial set of **45** items reproduced the expected 14 lower-level clusters of triplets and 3 higher-order metaclusters, with the exception of the UrSe (Urointestinal Sensitivity)..."

Lines 559-567:

"The final set of 33 items yielded a good fit (**TEFI = -77.23**), with all items showing high cluster stability ($> 90\%$), with the exception of Olfa. The final structure included 11 lower-level clusters, grouped into 3 higher-order metaclusters: "Interoceptive Deficit (containing CaCo, UrIn, Derm, Sati, Olfa), "Interoceptive Awareness" (containing ExAc, RelA, SexS), and "Visceroception" (containing Card, Resp, Gast). **Dimension stability for the lower clusters was generally very high, ranging from 100% (SexS, Card, Gast, Resp, RelA, ExAc) to 99.2% (Urin). Other dimensions also exhibited strong stability, with Derm and Sati at 99.6%, RelA at 99.8%, and CaCo at 99.4%. Additionally, two out of the three high-order clusters showed good**

dimension stability ("Interoceptive Awareness =96.6% , Visceroception = 99.4%), while Interoceptive Deficit showed poor dimension stability (42%)."

3. The design of study 1 supports the examination of structural invariance of their instrument; however, it's somewhat unclear what the authors hypotheses for why structural invariance would or would not hold are. Given that the analyses and results for structural invariance comprise a nontrivial amount of study 1, I would encourage the authors to provide some additional discussion about the potential impact of structural invariance in the context of interoception. Alternatively, the authors may wish to move this to supplemental materials.

We agree with the reviewer that the issue of structural invariance is primarily methodological and not unique to interception research. Questionnaire presentation and format (e.g., item grouping, ordering) can substantially influence the structure that is observed (Sahin 2021; Stefkovics & Kmetty, 2022), and we believe these factors warrant careful examination during instrument development.

Because our work aims to contribute to broader improvements in psychometric methodology, we are concerned that removing this section entirely would omit an important element of our scale development process.

At the same time, while we consider our original approach a useful experimental illustration of how such issues can be investigated, we acknowledge that it is not central to the paper's main contribution and as such if the reviewer truly believes it undermines the main priority of the paper we will move this to the supplementary materials.

4. The authors provide a helpful visualization of their modality-by-context approach in Figure 1; however, in examining the first graphic in Figure 1, some modality-facet combinations have considerably fewer initial items than others. What was the author's rationale for the number of initial items included here, particularly for combinations with only 1 initial item? Additionally, how do the authors think this approach may have impacted the resulting structure of the Mint Scale? Generally speaking, it is recommended to write more items than you expect to retain (e.g., Clark & Watson, 1995) so I'm a little concerned that modality-facet combinations that began with a very small number of item may be less likely to be retained in the final instrument. If this is the case, some discussion of this in the limitations would be warranted.

We believe our item generation strategy was consistent with Clark and Watson's (1995/2019) recommendations: the initial pool was intentionally larger than the anticipated final set to ensure broad content coverage. Unequal item counts across modality–facet combinations primarily reflect differences in how readily distinct, non-redundant indicators could be developed. For some domains, additional items tended to vary only superficially in wording rather than in substantive content; in these cases, we prioritized conceptual clarity and representational adequacy over item quantity.

We agree with the reviewer that combinations with fewer initial items may have had a lower probability of retention due to fewer opportunities for empirical optimization. However, we believe that even domains represented by fewer items were adequately covered conceptually, and in some cases single items were retained following careful analytical and theoretical evaluation of their role within the overall structure (e.g., "Sometimes I am not sure whether I need to go to the toilet or not [to urinate or defecate]").

We believe that we have tackled this issue adequately but have added to the limitation section that the current coverage of the Mint is not exhaustive despite best efforts (see above).

5. The authors do not provide information for how to score the scales and subscales of their instrument. I assume that simple sum scores or item averages are used but it would improve clarity if the authors specified their scoring algorithm.

We thank the reviewer for pointing out this foresight we have now added to the manuscript that we score averages for the metaclusters.

Lines 400-403:

The 45 items of the Multimodal Interoception Questionnaire (Mint) scale were presented in a random order, with the same 7-point Likert scale as in study 1 (0 = Disagree, 6 = Agree). **Scores should be calculated as the average of the items within each lower-order cluster. However, when a broader summary is required, average scores for the higher-order meta-clusters can be computed from the average of all their corresponding items.** Correspondingly, the high-order meta-clusters of the mint showed good to excellent internal consistency: Interoceptive Awareness ($\omega = 0.85$), Interoceptive Deficit ($\omega = 0.74$), Visceroception ($\omega = 0.88$)."

6. Although the authors provide reliability metrics for the psychometric networks used to develop their scale, they don't appear to have provided reliability metrics for the final scores from their measure. I encourage the authors to report reliability

metrics for their final scale scores for both studies.

We have now calculated internal consistency for the MINT and added the following to the manuscript.

Line 304:

"The final hierarchical EGA model (**TEFI**= -119.18), in which all 54 items yielded a high cluster stability (> 90%) suggested 3 metaclusters and 15 lower-level clusters (each containing 3 items): "Interoceptive Deficits" (containing 5 clusters: Urointestinal Inaccuracy - UrIn; Cardiorespiratory Confusion - CaCo; Cardiorespiratory Noticing - CaNo; Olfactory Compensation - Olfa; Satiety Noticing - Sati), "Interoceptive Awareness" (containing 7 clusters: Sexual Arousal Awareness - SexA; Sexual Arousal Sensitivity - SexS; Sexual Organs Sensitivity - SexO; Urointestinal Sensitivity - UrSe; Relaxation Awareness - RelA; State Specificity - StaS; Expulsion Accuracy - ExAc), and "Interoceptive Sensitivity" (containing 6 clusters: Cardioception - Card; Respiroception - Resp; Signalling - Sign; Gastroception - Gast; Dermal Hypersensitivity - Derm; Sexual Arousal Changes - SexC). "

Lines 403-405:

"The 45 items of the Multimodal Interoception Questionnaire (Mint) [...].

Correspondingly, the high-order meta-clusters of the mint showed good to excellent internal consistency: Interoceptive Awareness ($\omega = 0.85$), Interoceptive Deficit ($\omega = 0.74$), Visceroception ($\omega = 0.88$)."

Line 559:

The final set of 33 items yielded a good fit (**TEFI** = -77.23).

7. Relatedly, it would strengthen the manuscript if the authors provided descriptive statistics for their scale scores. I would also encourage the authors to assess for potential demographic differences in their scale score and score reliability (e.g., AERA, APA, & NCME, 2014).

We agree that including descriptive statistics would strengthen the manuscript. We have now computed means and standard deviations for all lower-level clusters and higher-order metaclusters of the final MINT scale and added them to the table below (Table 3 in the manuscript). Additionally, we calculated Cohen's *d* to estimate the magnitude of gender differences and assessed their statistical significance using independent-samples *t*-tests.

	Female	Male	Cohen's d
MINT - Visceroception	3.26 (1.20)	3.02 (1.18)	d = 0.20**
MINT - Cardioception	3.35 (1.48)	2.96 (1.39)	d = 0.27***
MINT - Respiroception	3.01 (1.44)	2.84 (1.42)	d = 0.12
MINT - Gastroception	3.42 (1.41)	3.28 (1.44)	d = 0.10
MINT - Deficit	3.04 (0.87)	2.68 (0.85)	d = 0.42***
MINT - Cardiorespiratory Confusion	2.57 (1.49)	1.80 (1.39)	d = 0.53***
MINT - Dermal Hypersensitivity	4.09 (1.24)	3.52 (1.32)	d = 0.45***
MINT - Urointestinal Inaccuracy	2.71 (1.44)	2.43 (1.37)	d = 0.20**
MINT - Satiety Noticing	3.15 (1.45)	2.97 (1.24)	d = 0.13
MINT - Olfactory Contribution	2.69 (1.45)	2.66 (1.46)	d = 0.02
MINT - Awareness	4.08 (0.81)	4.46 (0.72)	d = -0.49***
MINT - Sexual Sensibility	4.53 (1.25)	4.94 (0.94)	d = -0.37***
MINT - Expulsion Accuracy	4.27 (1.07)	4.63 (0.97)	d = -0.35***
MINT - Relaxation Awareness	3.46 (1.25)	3.82 (1.06)	d = -0.31***
Cohen's d. Significance: *** p < .001, ** p < .01, * p < .05.			

At the metacluster level, the largest gender difference emerged for Interoceptive Awareness, with males reporting significantly higher scores than females. Notably, this was the only metacluster for which males scored higher than females, and this pattern was consistent across all of its lower-level clusters (Sexual Sensibility, Expulsion Accuracy, and Relaxation Awareness).

In contrast, females reported higher scores than males on the Interoceptive Deficit metacluster and on Visceroception, although the latter effect was small. At the lower-cluster level, the largest gender differences were observed for Cardiorespiratory Confusion and Dermal Hypersensitivity, with females scoring significantly higher than males. Smaller but significant differences were also found for Urointestinal Inaccuracy and Cardioception. No significant gender differences were observed for Respiroception, Gastroception, Satiety Noticing, or Olfactory Contribution.

We also added the following to the manuscript (lines 518-519):

“We then computed the lower-level dimensions and the higher-level metaclusters’ scores by averaging their corresponding items. **Descriptive statistics were calculated for each cluster, and potential demographic differences were examined using independent-samples t-tests.** The convergent validity (...)”

Lines 567-570:

“Figure 7 presents gender differences across these clusters. The largest metacluster-level difference was observed for *Interoceptive Awareness*, whereas the largest lower-level difference was found for *Cardiorespiratory Confusion*. Item centrality (interpretable as loadings in traditional factor analysis frameworks) are shown in Figure 3.”

8. The authors chose to reduce the 42 items in study 1 to 33 items in study 2 based on conceptual rationale. This was somewhat unexpected as the purpose of study 2 was to validate the ultimate structure obtained in study 1; however, the removal of 9 items (~20%) based on study 2 data make the clean split between the development and validation studies a bit muddier. This may not be needed in the manuscript but some additional justification for this choice would be helpful for evaluating this design decision. Would the authors be able to provide more detail?

We agree with the reviewer that additional justification is warranted for the removal of items between Study 1 and Study 2. One motivating factor was the aim to produce a practically usable instrument; a 45-item scale was considered relatively lengthy, which prompted further evaluation of item redundancy, conceptual overlap, and face validity. This led us to refine the item pool prior to final validation. The following lines have been added to the manuscript to clarify our decision to further remove items from study 1 to study 2:

Line 552-558:

“The application of EGA to the initial set of 45 items reproduced the expected 14 lower-level clusters of triplets and 3 higher-order metaclusters, with the exception of the *UrSe* (Urointestinal Sensitivity), for which one item moved to the *Olfa* (Olfactory Compensation) cluster. **Given our aim to develop a practically usable instrument and to align its length with other widely used and conceptually related measures discussed here, we considered a 45-item scale to be overly lengthy. This prompted a further evaluation of item redundancy, conceptual overlap, and face validity (e.g., Urointestinal Sensitivity items showed instability, and Sexual Organs Sensitivity items overlapped with Sexual Arousal Awareness items), leading to a reduction of the item pool to 33 items.”**

As stated above, we performed EGA on a new independent sample to assess the replicability of the 33-item Mint. In summary, the factor structure on the new independent sample was largely replicated, with only a minor deviation: the Dermal Hypersensitivity dimension shifted from the Interoceptive Deficit meta-cluster to the Interoceptive Awareness cluster (review answer to editor above and lines xx–xx).

9. I recommend the authors include descriptive statistics and measures of reliability for the different score versions of the Mint scale used in study 2

We have reported descriptive statistics and measures of reliability for the different scores computed from the Mint questionnaire (see above for omega scores and table for descriptive statistics).

10. The number of correlational and regression analyses conducted in study 2 was somewhat overwhelming and difficult to follow. It would be helpful if the authors added subheadings to this section to help guide the reader through the analyses.

We have now added the same subheadings as in the method section to the result section where relevant.

11. I have some concerns with the reporting of the criterion validity results. The authors use both R^2 and BIC to determine the “best” predictor scores of each criterion variable however, I wasn’t able to find results which communicated how well each criterion was explained by each set of predictors scores. Its possible, for example, that the “best” model might only explain 1% of the outcome variability which might not be particularly meaningful from a practical perspective. Providing overall model fit, either in the manuscript or as supplemental material, would strengthen the interpretations of the regression analysis.

We have now added information in the Results section on the R^2 of the best-performing model. To keep the Results concise, we direct readers to the Supplementary Materials, where the R^2 values for all tested models are reported. The following has been added to the results section (lines 648-658):

“Most of the target outcomes measured in the study to assess validity were best predicted by one of the Mint dimension (see Figure 5). **For example, Deficit was the strongest predictor for several outcomes, including Alexithymia ($R^2 = 20-22\%$), ERS’ Arousal ($R^2 = 20\%$), ADHD and Autism ($R^2 = 3\%$ each), Self and Reality ($R^2 = 21\%$), and somatic symptoms ($R^2 = 3-7\%$). CaCo best predicted CEFSAs anomalous experiences of the body ($R^2 = 23\%$), while Sati was the strongest predictor for BMI ($R^2 = 3\%$). A few outcomes were better predicted by MAIA dimensions. In particular, Not-worrying predicted ERS’ Sensitivity ($R^2 = 22\%$) and Persistence ($R^2 = 20\%$), whereas Trusting was the strongest predictor for Life Satisfaction ($R^2 = 28\%$), Depression ($R^2 = 21\%$), Anxiety ($R^2 = 24\%$), CEFSAs Emotion ($R^2 = 17\%$) and Connection ($R^2 = 25\%$), as well as self-reported physical activity ($R^2 = 3\%$). Remaining model fits can be reviewed in the data analysis folder project's GitHub repository. “**

12. In study 2, I would encourage the authors to include interval estimates around their pearson's r values prior to comparing the raw values of the estimates as different (e.g., line 538 – line 541)

We have now added interval estimates for all pearsons'r values we present in the manuscript. Notably, we avoided adding this information to the results section for correlations represented in Figure 4 which shows only coefficients for significant correlations ($p < .001$).

Lines 624-628:

The BPQ was positively correlated with most Mint dimensions (the highest with *Visceroception*, $r=.46$, **95% CI = [0.40, 0.51]**). The IAS was positively correlated with *Visceroception* and *Awareness* dimensions (the highest with *Awareness*, $r=.63$, **95% CI = [0.58, 0.67]**), but negatively with most *Deficit* dimensions (with the exception of *Olf* and *Derm*). In turn, the *Visceroception* metacluster most strongly correlated with MAIA's Noticing ($r = .55$, **95% CI = [0.50, 0.60]**).

Lines 676-682:

Finally, the importance of heart monitoring (for owners of such wearable) was best predicted by MAIA's Body listening ($r=.52$, **95% CI= [0.35, 0.68]**), followed the *Card* dimension of the Mint ($r=.37$, **95% CI = [0.23, 0.51]**). The importance of sleep monitoring was best predicted by *RelA* ($r = .33$, **95% CI = [0.11, 0.56]**), followed by MAIA's not worrying ($r = -.35$, **95% CI = [-0.58, -0.11]**). Daily activity via steps monitoring was best predicted by MAIA's Noticing ($r=.50$, **95% CI = [0.32, 0.67]**), followed by *Awareness* ($r=.60$, **95% CI = [0.37, 0.83]**).

13. The authors use the area under the ROC curve to evaluate predictive performance of the various interoception measures for mental health outcomes. This measure can be sensitive to imbalance in the class distribution. It is difficult to evaluate the appropriateness of the ROC AUC (relative to the precision-recall AUC for example) without knowing the descriptive statistics for the mental health outcomes.

We thank the reviewer for raising the class imbalance point, which is an important consideration in predictive machine learning models. However, our use of logistic regression is inferential rather than predictive. We are not applying a threshold to classify observations (where class imbalance skews accuracy metrics), but rather estimating the coefficients to understand the relationships between predictors and the outcome.

It is an established mathematical property of logistic regression via Maximum Likelihood Estimation that while the intercept does reflect the sample base rate, the estimated coefficients (the slopes) and their resulting odds ratios are invariant to the ratio of class imbalance. Because our minority class contains a decent absolute number of observations, we believe our estimates of likelihood are stable and valid for our experimental design.

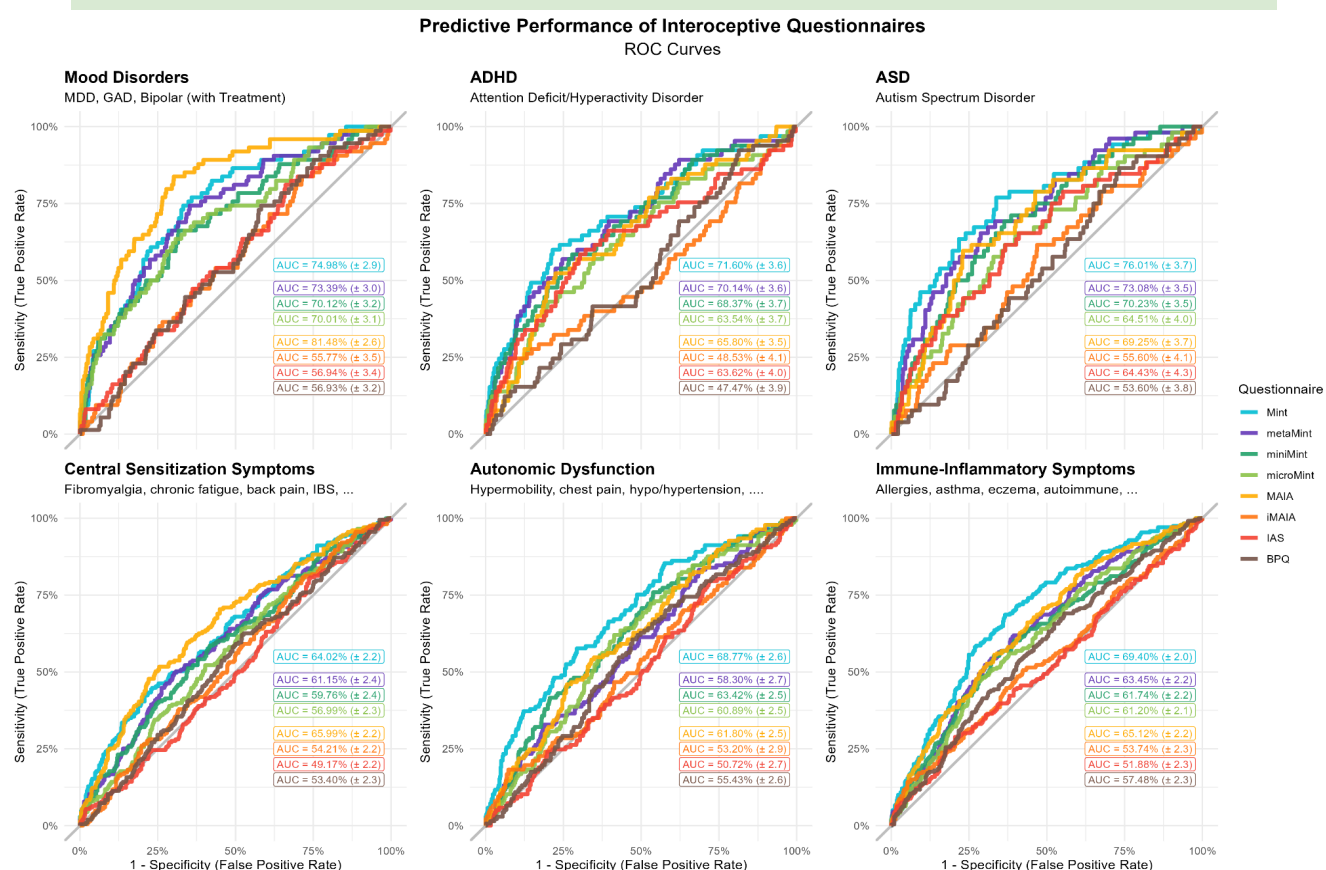
14. Additionally, I would encourage the authors to report bootstrapped standard errors around their AUC metrics to further support their claims about the Mint scale outperforming other scales. Standard errors would provide evidence that differences in AUC are not due to sampling variability alone.

We have now computed standard errors for the AUC metrics and added them to Figure 6. Specifically, 95% confidence intervals for the AUC were estimated using nonparametric bootstrap resampling with 1,000 iterations. The standard errors were then derived from the confidence interval widths using the formula

$$SE = \frac{Upper\ CI - Lower\ CI}{2 * 1.96}$$

which relies on the normal approximation to the sampling distribution of the AUC. This approach allows us to quantify the uncertainty around the AUC estimates and supports the comparison of performance between the Mint scale and other scales.

The figure looks like this following these changes:



15. The authors rely heavily on network psychometrics in their analyses and interpretation of the Mint scale without exploring any latent variable models. Prior work has recommended that latent variable models are examined prior to network psychometrics to reduce the chance of capturing and emphasizing spurious item correlations (Hallquist et al., 2021). Given the large number of correlational analyses in this paper, I would encourage the authors to explore the dimensionality of their instrument using traditional factor analytic methods and comment on any similarities or discrepancies.

We agree that discussing the similarities and differences between the network-based solution and traditional latent variable models would strengthen the manuscript. A summary of these analyses is provided below:

Within the factor-analytic framework, factor retention using oblimin rotation indicated both a 1-factor and a 17-factor solution for the original 120-item pool. The single-factor solution accounted for 15.67% of the variance ($\chi^2 = 23,694.21$, RMSEA = 0.07 [0.06, 0.07], TLI = 0.35). In contrast, the 17-factor model (oblimin rotation with polychoric correlations) demonstrated improved fit ($\chi^2 = 14,152.21$, RMSEA = 0.06 [0.05, 0.06], TLI = 0.68) and explained 52.23% of the total variance. However, the variance was distributed across many factors, with individual factors accounting for relatively small proportions (1.56%–5.14%), reflecting a highly multidimensional structure. Given the superior fit of the 17-factor model, this solution was retained.

Removal of items followed the same decision criteria applied in the EGA: 1) removal of items with primary loadings < .40 (31 items), 2) removal of items with salient cross-loadings (1 item), 3) exclusion of factors defined by only two items (4 items), and 4) retention of the three highest-loading items per factor where applicable (39 items). The final solution resulted in 44 items. We then performed one more fa on the 45 item solution ($\chi^2 = 1101.05$, RMSEA = 0.05 [0.05, 0.06], TLI = 0.86).

In terms of differences between the EGAMint and the FAMint, the exploratory factor analysis showed substantial convergence with the EGA structure. Several factors corresponded closely to EGA clusters, including gastric sensitivity, respiratory sensitivity, dermal hypersensitivity, urointestinal sensitivity, and sexual organ sensitivity. However, some discrepancies emerged. In particular, cardiorespiratory items were distributed across multiple EFA factors rather than forming distinct clusters as observed in the EGA solution. Additionally, accuracy and confusion items were sometimes combined within the same factor, whereas EGA separated these

constructs. Overall, the EFA solution broadly replicated the domain structure identified by EGA but yielded slightly broader factors.

We have added the following on the manuscript (lines 285-288):

“Lastly, we will assess model fit using the Generalized Total Entropy Fit Index (TEFI; Golino et al., 2021), where higher values (i.e., values closer to zero) indicate better fit. **We additionally conducted an exploratory factor analysis (EFA) following the same item-removal procedure and compared the resulting structure with the EGA solution to complement the psychometric network approach with a traditional factor-analytic framework.**”

Lines 315-332:

“The final hierarchical EGA model [...] and “Interceptive Sensitivity” (containing 6 clusters: Cardioception - Card; Respiroception - Resp; Signalling - Sign; Gastroception - Gast; Dermal Hypersensitivity - Derm; Sexual Arousal Changes - SexC). **Within the factor-analytic framework, factor retention using oblimin rotation indicated both a 1-factor and a 17-factor solution for the original 120-item pool. The single-factor solution explained 15.67% of the variance ($\chi^2 = 23,694.21$, RMSEA = 0.07 [0.06, 0.07], TLI = 0.35), whereas the 17-factor solution showed improved fit ($\chi^2 = 14,152.21$, RMSEA = 0.06 [0.05, 0.06], TLI = 0.68) and accounted for 52.23% of the total variance, supporting a multidimensional structure. Item reduction followed the same criteria applied in the EGA, and this procedure resulted in a final set of 44 items. A subsequent factor analysis of the reduced item pool showed acceptable model fit ($\chi^2 = 1101.05$, RMSEA = 0.05 [0.05, 0.06], TLI = 0.86). Overall, the EFA solution showed substantial convergence with the EGA structure, with several factors corresponding closely to clusters identified by EGA (e.g., gastric sensitivity, respiratory sensitivity, dermal hypersensitivity, urointestinal sensitivity, and sexual organ sensitivity). However, some differences emerged, particularly in the cardiorespiratory domain, where items were distributed across multiple factors in EFA rather than forming distinct clusters, and in the treatment of accuracy and confusion items, which were occasionally combined in EFA but separated in the EGA solution. Together, these results suggest that while the factor-analytic solution broadly replicated the domain structure identified by EGA, it tended to yield slightly broader factors.**”

Lastly, we conducted factor analysis on the final 33 items of the mint. For factor retention, we applied FA with oblimin rotation, and for examining factor structure, we used oblimin rotation with polychoric correlations. The optimal two-factor solution explained 30.63% of the variance ($\chi^2 = 5247.37$, RMSEA = 0.12 [0.12, 0.12], TLI = 0.50), indicating poor model fit. Consequently, we examined the next-best solution with nine factors, which showed improved fit, though still not ideal ($\chi^2 = 1053.59$, RMSEA = 0.06 [0.06, 0.07], TLI = 0.86). This nine-factor solution explained 56.88% of

the total variance, comprising eight factors with three items each (Gastrioception, Sexual Sensibility, Urointestinal Inaccuracy, Dermal Hypersensitivity, Expulsion Accuracy, Olfactory Compensation, Satiety Noticing, and Relaxation Awareness) and one factor with six items encompassing Cardioception, Respiroception, and Cardiorespiratory Confusion. Notably, this organisation is similar to the lower clusters of the final 33 item EGAMint, with the exception that all cardiorespiratory items were grouped together.

We added the following to the manuscript:

Lines 593-605:

“Additionally, we applied hierarchical clustering analysis which replicated this structure (see Figure 2), suggesting consistency across methods. **Lastly, to complement EGA we performed factor analysis on the final set of 33 items. Factor retention was examined using FA with oblimin rotation and polychoric correlations. The optimal 2-factor solution showed poor fit ($\chi^2 = 5247.37$, RMSEA = 0.12 [0.12, 0.12], TLI = 0.50) and explained only 30% of the variance. We therefore examined the next-best solution with nine factors, which demonstrated improved, though still suboptimal, fit ($\chi^2 = 1053.59$, RMSEA = 0.06 [0.06, 0.07], TLI = 0.86) and explained 56.88% of the total variance. This solution comprised eight factors with three items each (Gastrioception, Sexual Sensibility, Urointestinal Inaccuracy, Dermal Hypersensitivity, Expulsion Accuracy, Olfactory Compensation, Satiety Noticing, and Relaxation Awareness) and one six-item factor combining Cardioception, Respiroception, and Cardiorespiratory Confusion. Overall, this structure closely resembled the lower-level clusters identified in the final 33-item EGA solution, with the main difference being that all cardiorespiratory items loaded onto a single factor rather than forming separate clusters. ”**

Reviewer 3

Comments to the Author

The authors present a series of studies that introduce the development and validation of the Multimodal Interoception Questionnaire (Mint scale) for measuring interoception. Strengths of the manuscript include the use of two samples to develop and validate the measure, and a strong focus on modeling the context and full range of modalities to extend and improve upon existing interoception measures. However, the limited rationale for the use of network analyses (as compared to factor analyses) prevented me from properly interpreting and evaluating the data and implications from this study.

We thank Reviewer 3 for recognizing the strengths of our study, particularly our focus on modeling context and the full range of modalities. We greatly appreciate your constructive comments, which have helped us better clarify our rationale for using network psychometrics and significantly improved the manuscript's readability.

The authors reference the innovation in using network psychometric methods multiple times in the manuscript; however, their rationale for using this method is unclear. I recommend the authors explain the justification for using network psychometric methods and how these methods differ from factor analytic methods (particularly as the Assessment readership may be more familiar with factor analytic methods).

We agree that the rationale for using network psychometric methods should be clarified, particularly for readers more familiar with factor analytic approaches.

The decision to use EGA was motivated by both conceptual and methodological considerations. The item pool was intentionally designed to span multiple physiological domains (e.g., cardiac, respiratory, genital) and multiple contexts and modalities (e.g., positive and negative arousal, sensitivity, accuracy), making cross-domain associations and partially overlapping dimensions likely.

Traditional approaches, such as EFA, assume that covariance among items is generated by latent variables that account for shared variance. In complex structures characterised by high inter-factor correlations or cross-loadings, however, global covariance patterns can dominate, potentially biasing the estimation of the number of dimensions (Christensen & Golino, 2021b).

In contrast, EGA estimates a network of conditional associations among items and identifies dimensions as communities of strongly connected nodes (Golino & Epskamp, 2017). That is, rather than attributing associations to latent causes, this approach allows structure to emerge from patterns of direct relationships among items. We believe this framework would be particularly useful when constructs may reflect interacting components (e.g., bodily sensations that co-activate across systems such as heart rate and breathing increasing at the same time) rather than a single underlying trait.

On a methodological level, simulation studies indicate that EGA performs comparably to or better than traditional factor retention methods in complex structures, including bi-factor structures and ordinal data (Jiménez et al., 2023; Markos & Tsigilis, 2024).

For these reasons we believed EGA was the most appropriate method and have now clarified this in the manuscript on lines 250-266:

“The structure of the items was analyzed using the recently developed Exploratory Graph Analysis (EGA, H. F. Golino & Epskamp, 2017) framework, which allows to jointly estimate the number of dimensions (i.e., clusters of items), the structure, as well as its stability using bootstrapping (H. Golino et al., 2020). At a fundamental level, EGA conceptualizes variables as nodes in a network, with connections (edges) reflecting associations between them. **We selected this approach for both conceptual and methodological reasons. The item pool was intentionally heterogeneous, spanning multiple physiological domains and psychological contexts. We therefore anticipated cross-domain associations and potential correlated or partially overlapping dimensions. Traditional factor analytic models explained this covariance among items by assuming latent variables that generate shared variance. However, in complex structures characterised by high inter-factor correlations or low cross-loadings, global variance patterns can dominate, potentially biasing the estimating of the number of dimensions (Christensen & Golino, 2021b). In contrast, EGA estimates a network of conditional associations among items and identifies dimensions as communities of strongly connected nodes (Golino & Epskamp, 2017). Rather than attributing associations to latent causes, this approach allows structure to emerge from patterns of direct relationships among items. This framework is particularly appropriate when constructs may reflect interacting components (e.g., bodily sensations that co-activate across systems such as heart rate and breathing increasing at the same time) rather than purely reflective latent traits. Moreover, simulation studies indicate that EGA performs comparably to or better than traditional factor retention methods in complex and highly correlated structures (Christensen & Golino, 2021b; Jiménez et al., 2023).**”

We do believe and agree with previous comments from the reviewers that adding a discussion on the differences and similarities between the EGA and FA results would strengthen our manuscript. We made changes to lines xx-xx and xx-xx to tackle this point (see response above).

I also recommend the authors provide background information on psychometric network analyses in the Introduction to help the reader understand this method.

We have now added more information on the introduction about network psychometrics to help readers understand this method and why we chose it over more traditional ones.

Lines 122–135:

“In Study 1, we describe the development of the Mint, using a “modality-by-context” framework to generate a comprehensive item pool, which was then refined using cuttingedge psychometric network analysis. **Network psychometrics is an emerging framework for studying relationships among psychological variables by representing them as a network structure (Golino & Epskamp , 2017). In this approach, psychological variables (e.g., traits or questionnaire items) are modelled as nodes, while the statistical relationships between them are represented as edges connecting the nodes. These edges can be weighted to reflect the strength and direction of associations between variables. Beyond descriptive modelling, network psychometrics has increasingly been applied to scale development and dimensionality assessment (Golino & Demetriou, 2017; Christensen et al., 2020; Magnavita & Chiorri, 2022; Wongvorachan et al., 2025). When analysing psychological measurement data, a central question concerns whether observed variables organise into groupings that reflect underlying constructs. Within a latent variable framework, items measuring the same construct are expected to show stronger mutual associations, resulting in cohesive clusters within a network representation. Accordingly, clusters of interconnected nodes in psychometric networks have been proposed to correspond to latent variables (Golino & Epskamp, 2016).”**

Relatedly, as network analyses have been used for over a decade now and have generated some criticism in the psychology community, I encourage the authors to consider tempering the language of some references to network analyses that could be seen as using boosterism language (e.g., cutting-edge psychometric network-based techniques; page 7 line 132).

We have now removed all mentions of “cutting-edge” from the manuscript.

It would be helpful to review a brief description of the attention check items

We agree with the reviewers that a brief description of the attention checks would be useful for readers. Hence we have added the following to the manuscript to clarify the structure of the attention checks in study 1 (lines 216-221):

“We additionally crafted 8 “attention check” items blending in (and distributed across) each category. **These items were designed to match the grammatical structure and response format of the surrounding items, in order to minimise their salience and reduce the risk of demand characteristics. Each attention check explicitly instructed participants to select a specific response option (e.g., the lowest, leftmost, or rightmost point on the scale), thereby allowing us to assess**

whether participants were attentively reading and following item instructions (see items in green in initial item list for more information). “

To clarify the attention checks used in Study 2 the following has been added to the manuscript.

Lines 506-512:

“7 attention checks were embedded throughout (one within each major questionnaire). **Specifically, the following items were used: “I can always accurately answer to the extreme left on this question to show that I am reading it” (Mint); “I notice that I am being asked to respond all the way to the right” (MAIA-2); “I can always accurately choose the lowest option” (IAS); “Respond all the way to the right” (BPQ-VSF); “I am able to respond all the way to the left” (TAS-20); “On the whole, I know I must press the highest option” (PI-18); and “I feel that to show I am being attentive I will press the lowest option” (CEFSA-14).”**

Lastly, some of the paragraphs in the manuscript are very long which decreases readability. I recommend the authors consider reorganizing text into paragraphs of 100-200 words in length.

We have revised the manuscript to shorten and restructure the paragraphs such that none exceeds 200 words. We believe this improves the overall readability and clarity of the manuscript.

I appreciate the authors’ efforts to strengthen the field of interoceptive assessment and research and hope these comments are helpful in preparing the manuscript further.

We thank the reviewer for acknowledging our efforts to advance the field of interoceptive assessment. The reviewer’s comments were constructive and helpful, and we believe that addressing them has substantially improved the manuscript.